Data Management Plan

Coastal observations: reproducible data management

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Research overview

Project Number	COAST-2026
Start date	2026-01-01
End date	2027-12-31

Synthetic scenario for testing all fifteen output questions. New coastal temperature measurements will be compared with two open reference datasets. This is not an approved or complete DMP.

People and responsibilities

- **Example research team**
  research@example.org
  Roles: Contact Person, Creator of DMP, Data Steward

Knowledge model: Common DSW Knowledge Model, 2.7.0 (dsw:root:2.7.0)

Generated with Data Stewardship Wizard.

Section 1: Data description and collection or re-use of existing data

1. How will new data be collected or produced and/or how will existing data be re-used?

Instrument datasets

- **Coastal temperature measurements**

  For this dataset, we are using the following instruments:

  - **Temperature logger**
    – Record temperature every ten minutes; retain UTC timestamps and calibration records.

  Quality control measures for **Coastal temperature measurements**: measurement calibration, data entry validation.

Re-used datasets

Open coastal observations — station 1

Source: https://example.org/datasets/coastal-observations/station-1/version-2026

Compare seasonal changes in water temperature.

- Retain the original timestamps.
- Document any excluded observations.

Version	2026.1

If a new version becomes available during the project, the current version will continue to be used.

Open coastal observations — station 2

Source: https://example.org/datasets/coastal-observations/station-2/version-2026

Compare seasonal changes in water temperature.

- Retain the original timestamps.
- Document any excluded observations.

Version	2026.1

If a new version becomes available during the project, the current version will continue to be used.

Shared arrangements

The following arrangements apply to all reference datasets selected for re-use above.

We will retain a copy of this dataset and provide it alongside the research results. This dataset is freely available with an obligation to cite the source. The data can be used directly without format conversion.

Provenance

Record source versions and processing scripts in the project repository.

Record	Retention
Processing log	10 years
Source checksum	10 years

2. What data (for example the kinds, formats, and volumes) will be collected or produced?

Instrument datasets

Data collection for **Coastal temperature measurements**: This dataset will be collected by experts in the project, with our own equipment. The equipment is very well described and known.

Data formats and types

Data format: **CSV (UTF-8)**. It is a standardized format suitable for long-term archiving. The estimated data volume is 120 GB.

Section 2: Documentation and data quality

3. What metadata and documentation (for example the methodology of data collection and way of organising data) will accompany the data?

We will be documenting the data with Dublin Core metadata standard. Metadata will be openly available including instructions how to get access to the data. Metadata will be available in a form that can be harvested and indexed (managed by the used repository / repositories).

Data storage and file conventions

We estimate the storage space that the project will require for all data and software (including temporary storage) to be 2048 gigabytes. We will use a filesystem with files and folders. We document how we manage file versioning for files and folders.

File naming conventions:

Name files station_YYYYMMDD.csv and document corrections in CHANGELOG.md.

4. What data quality control measures will be used?

Quality control measures for **Coastal temperature measurements**: measurement calibration, data entry validation.

Section 3: Storage and backup during research process

5. How will data and metadata be stored and backed up during the research process?

There is a shared workspace used during the project for working with data. Dedicated specialists will operate the shared workspace. Storage needs will be the same during the whole project. All

essential data will also be stored elsewhere to prevent a total loss of data. We will make automated backups of all data stored outside the workspace.

Archival during the project

Data will be archived in cold storage during the project. The archive will be stored on tape. The archive will be stored at a remote location.

Changes to archived data will require re-archival. Frequent backups are required because the archived data changes frequently. These backups will be used to recover from accidental changes or deletions.

6. How will data security and protection of sensitive data be taken care of during the research?

Project members will not carry data with them (e.g. on laptops, USB sticks, or other external media). All project web services are accessible via secure HTTP (https://...). Project members have been instructed about both generic and specific risks to the project.

See question 5 for archival and backup arrangements.

Section 4: Legal and ethical requirements, codes of conduct

7. If personal data are processed, how will compliance with legislation on personal data and on security be ensured?

We will not collect any data related to individuals.

8. How will other legal issues, such as intellectual property rights and ownership, be managed? What legislation is applicable?

For the reference and non-reference data sets that we reuse, conditions are as follows:

- Open coastal observations — station 1
  It is freely available with obligation to quote the source (e.g. CC-BY).
- Open coastal observations — station 2
  It is freely available with obligation to quote the source (e.g. CC-BY).

9. What ethical issues and codes of conduct are there, and how will they be taken into account?

Ethical approvals

The questionnaire records the following ethical approval arrangements:

- **Coastal observations: reproducible data management**
  According to the questionnaire, this project does not require ethical approval.

Data we produce

For the data we produce, the ethical aspects are as follows:

- **Quality-checked coastal observations**
  - It does not contain personal data.
  - It does not contain sensitive data.
- **Internal instrument checks**
  - no information on personal nor sensitive data.

Data we collect

We will not collect any data related to individuals, i.e. "personal data".

The data collection is not subject to ethical legislation.

Section 5: Data sharing and long-term preservation

10. How and when will data be shared? Are there possible restrictions to data sharing or embargo reasons?

Quality-checked coastal observations

This dataset will be published.

Distribution 1

Open access: this distribution will be shared with anyone. This distribution will be deposited in an institutional repository. Licence start date: 2027-12-31. Freely available with obligation to quote the source (e.g. CC-BY).

Distribution 2

Access will be limited to a predefined group of people. This distribution will be deposited in a special-purpose repository for the project.

Licence start date: 2027-06-01. Access to this distribution is subject to restrictions.

This preliminary release is limited to the project review group.

- Do not redistribute the preliminary files.
- Cite the release identifier in review notes.

Re-users will be able to get access by request to the contact person. The conditions will be published as part of open metadata. Restriction details: https://example.org/coast/review-terms.

Internal instrument checks

This dataset will not be published.

11. How will data for preservation be selected, and where will data be preserved long-term (for example a data repository or archive)?

Quality-checked coastal observations

Quality-checked coastal measurements with a station dictionary.

Keep Selection-2027-12-31.csv alongside the processing record.

This dataset contains processed data leading to published results. This dataset will be published. Retention period (prepaid): 10 years. The metadata will be available even when the data no longer exists.

Related paper: https://example.org/papers/coast-study

Preservation destinations by distribution:

- **Distribution 1:** Our institutional repository.
- **Distribution 2:** Special-purpose repository for the project. Whether we will be able to support this repository for a sufficiently long time has not been specified.

Internal instrument checks

Instrument test records retained for internal verification.

This dataset contains intermediate data. This dataset will not be published.
Other reasons for not publishing this dataset:

The instrument supplier restricts publication of test records.

Internal verification copies will still be kept.

- Record the access decision.
- Retain the calibration context.

Retention period (prepaid): 15 years. The metadata will be available even when the data no longer exists.

Repository costs and publication preparation

We have budgeted for the costs in the project to pay charges of the used repositories. We have a reserved budget for the time and effort it will take to prepare the data for publication.

Project-wide cold storage after the project

The project will keep copies of its data in cold storage after it finishes. Departmental budgets of one or more project participants will pay for this long-term storage. The archive will be kept for at least 12 years. The archival period can be extended. Library or archive staff will decide on an extension. Basis for extending the archival period: actual use of the archived data, predicted use of the archived data, available budget.

Obsolete data formats in cold storage will be upgraded. Archived data will not be migrated regularly to more modern storage media.

12. What methods or software tools are needed to access and use data?

Quality-checked coastal observations

Software required to use this dataset:

- **CoastView 1.0 (synthetic test tool)**
  Available at: https://example.org/coastview/1.0

Internal instrument checks

This dataset will not be published.

13. How will the application of a unique and persistent identifier (such as a Digital Object Identifier (DOI)) to each data set be ensured?

Quality-checked coastal observations

Distribution 1 Institutional repository

The repository will assign the persistent identifier. The repository will make sure the persistent identifier can be resolved to a digital object.

Distribution 2 Special-purpose repository

An institutional data steward will assign the persistent identifier. The repository will make sure the persistent identifier can be resolved to a digital object.

Internal instrument checks

This dataset will not be published.

Possible re-use of created data in other contexts

- **Coastal temperature measurements**
  Other researchers in this field will be interested in re-using this data.

Section 6: Data management responsibilities and resources

14. Who (for example role, position, and institution) will be responsible for data management (i.e. the data steward)?

Example research team is responsible for the management and proficiency of data including data processing, data policies, data guidelines, and data availability.

Example research team is responsible for implementing the DMP, and ensuring it is reviewed and revised.

15. What resources (for example financial and time) will be dedicated to data management and ensuring that data will be FAIR (Findable, Accessible, Interoperable, Re-usable)?

Additional specialist expertise is required. Existing staff will receive training.

Staff will complete training before data publication.

- Metadata quality checks
- Version control and provenance

We do not require any hardware or software in addition to what is usually available in the institute. The selected repositories charge for their services.

Data-management budget

Resource and purpose	Budget	Funding source
Extended resource 9	900 TWD	Institutional research funds
Prepare metadata and validate deposit files. MIX-LONG-09-PARA-01: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-02: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-03: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-04: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-05: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-06: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-07: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-08: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-09: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-10: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-11: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-12: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-13: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-14: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-15: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-16: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-17: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-18: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-19: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-20: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-21: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-22: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-23: Retain the original resource justification and its audit trail. MIX-LONG-09-PARA-24: Retain the original resource justification and its audit trail.		

Resource and purpose	Budget	Funding source
Extended resource 9	900 TWD	Institutional research funds

MIX-LONG-09-PARA-25: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-26: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-27: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-28: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-29: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-30: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-31: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-32: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-33: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-34: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-35: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-36: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-37: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-38: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-39: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-40: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-41: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-42: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-43: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-44: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-45: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-46: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-47: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-48: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-49: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-50: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-51: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-52: Retain the original resource justification and its audit trail.
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MIX-LONG-09-PARA-55: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-56: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-57: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-58: Retain the original resource justification and its audit trail.
MIX-LONG-09-PARA-59: Retain the original resource justification and its audit trail.

Resource and purpose	Budget	Funding source
Extended resource 9	900 TWD	Institutional research funds
MIX-LONG-09-PARA-60: Retain the original resource justification and its audit trail. • Preserve Budget-2027-12-31.csv. • Keep the funding record. This resource supports findability and management of data.		

Resource and purpose	Budget	Funding source
Data preparation resource 1 Prepare metadata and validate deposit files. This resource supports findability and management of data.	5000 TWD	Institutional research funds
Data preparation resource 2 Prepare metadata and validate deposit files. This resource supports findability and management of data.	0 TWD	Institutional research funds
Budget resource 3 Prepare metadata and validate deposit files. This resource supports findability and management of data.	300 TWD	Institutional research funds
Budget resource 4 Prepare metadata and validate deposit files. This resource supports findability and management of data.	400 TWD	Institutional research funds
Budget resource 5 Prepare metadata and validate deposit files. This resource supports findability and management of data.	500 TWD	Institutional research funds
Budget resource 6 Prepare metadata and validate deposit files. This resource supports findability and management of data.	600 TWD	Institutional research funds

Resource and purpose	Budget	Funding source
Budget resource 7 Prepare metadata and validate deposit files. This resource supports findability and management of data.	700 TWD	Institutional research funds
Budget resource 8 Prepare metadata and validate deposit files. This resource supports findability and management of data.	800 TWD	Institutional research funds
Boundary resource 10 Prepare metadata and validate deposit files. This resource supports findability and management of data.	1000 TWD	Institutional research funds
Boundary resource 11 Prepare metadata and validate deposit files. This resource supports findability and management of data.	1100 TWD	Institutional research funds
Boundary resource 12 Prepare metadata and validate deposit files. This resource supports findability and management of data.	1200 TWD	Institutional research funds
Boundary resource 13 Prepare metadata and validate deposit files. This resource supports findability and management of data.	1300 TWD	Institutional research funds
Boundary resource 14 Prepare metadata and validate deposit files. This resource supports findability and management of data.	1400 TWD	Institutional research funds
Boundary resource 15 Prepare metadata and validate deposit files. This resource supports findability and management of data.	1500 TWD	Institutional research funds

Resource and purpose	Budget	Funding source
Boundary resource 16 Prepare metadata and validate deposit files. This resource supports findability and management of data.	1600 TWD	Institutional research funds
Boundary resource 17 Prepare metadata and validate deposit files. This resource supports findability and management of data.	1700 TWD	Institutional research funds
Boundary resource 18 Prepare metadata and validate deposit files. This resource supports findability and management of data.	1800 TWD	Institutional research funds
Boundary resource 19 Prepare metadata and validate deposit files. This resource supports findability and management of data.	1900 TWD	Institutional research funds
Boundary resource 20 Prepare metadata and validate deposit files. This resource supports findability and management of data.	2000 TWD	Institutional research funds
Boundary resource 21 Prepare metadata and validate deposit files. This resource supports findability and management of data.	2100 TWD	Institutional research funds
Boundary resource 22 Prepare metadata and validate deposit files. This resource supports findability and management of data.	2200 TWD	Institutional research funds
Boundary resource 23 Prepare metadata and validate deposit files. This resource supports findability and management of data.	2300 TWD	Institutional research funds

Resource and purpose	Budget	Funding source
Boundary resource 24 Prepare metadata and validate deposit files. This resource supports findability and management of data.	2400 TWD	Institutional research funds
Boundary resource 25 Prepare metadata and validate deposit files. This resource supports findability and management of data.	2500 TWD	Institutional research funds
Boundary resource 26 Prepare metadata and validate deposit files. This resource supports findability and management of data.	2600 TWD	Institutional research funds
Boundary resource 27 Prepare metadata and validate deposit files. This resource supports findability and management of data.	2700 TWD	Institutional research funds
Boundary resource 28 Prepare metadata and validate deposit files. This resource supports findability and management of data.	2800 TWD	Institutional research funds
Boundary resource 29 Prepare metadata and validate deposit files. This resource supports findability and management of data.	2900 TWD	Institutional research funds
Boundary resource 30 Prepare metadata and validate deposit files. This resource supports findability and management of data.	3000 TWD	Institutional research funds
Boundary resource 31 Prepare metadata and validate deposit files. This resource supports findability and management of data.	3100 TWD	Institutional research funds

Resource and purpose	Budget	Funding source
Boundary resource 32 Prepare metadata and validate deposit files. This resource supports findability and management of data.	3200 TWD	Institutional research funds